

FinRelief AI

AI-Powered Debt Relief & Financial Recovery Platform

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND TECHNOLOGY

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1. Project Title

FinRelief AI – AI-Powered Debt Relief & Financial Recovery Platform

2. Abstract

FinRelief AI is a web application that helps users analyze their financial health, manage debts, predict loan settlements, and receive AI-powered financial recommendations. The system enables users to monitor their income, expenses, loans, and financial stress through an interactive dashboard.

3. Objective

The main objectives are:

- Help users monitor financial health.
 - Calculate debt-to-income ratio.
 - Analyze EMI burden.
 - Predict loan settlement percentage.
 - Generate negotiation letters.
 - Provide AI-based financial advice.
 - Improve financial planning.
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4. Technologies Used

Frontend

- React.js
- Vite
- Axios
- CSS3

Backend

- FastAPI
- Python 3.11
- SQLAlchemy
- Pydantic
- JWT Authentication
- Uvicorn

Database

- SQLite (Local Development)
- PostgreSQL (Render Deployment)

Deployment

- GitHub
 - Render
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5. Software Requirements

- Windows 10/11
 - VS Code
 - Python 3.11
 - Node.js
 - Git
 - Render Account
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6. Hardware Requirements

- Intel i3/i5 Processor
 - 4 GB RAM (Minimum)
 - 10 GB Free Storage
 - Internet Connection
-

7. Modules

User Authentication

- Registration
- Login
- JWT Authentication

Dashboard

Displays

- Monthly Income
- Monthly Expenses
- Monthly Surplus
- Total EMI

- Outstanding Debt
- Financial Health Score

Financial Health

Calculates

- EMI Ratio
- Debt-to-Income Ratio
- Monthly Surplus
- Financial Stress Level

Settlement Predictor

Predicts

- Settlement Percentage
- Settlement Amount
- Recovery Suggestions

Negotiation Letter

Generates professional loan negotiation letters.

Loan History

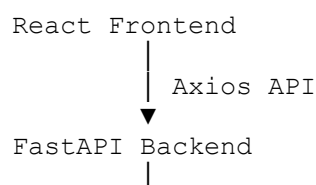
Shows

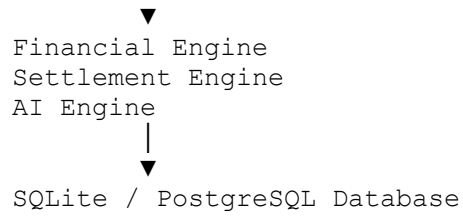
- Added Loans
- Loan Status
- Payment Details

Know Your Rights

Provides debt-related legal information and consumer rights.

8. System Architecture





9. Database Tables

Users

- user_id
- username
- email
- password

Loans

- loan_id
- user_id
- lender
- loan_amount
- outstanding_amount
- interest_rate
- emi
- due_date

Financial Profile

- profile_id
- user_id
- monthly_income
- monthly_expenses
- existing_debts
- lump_sum_available
- financial_health_score

10. Features

- Secure Login
- Loan Management

- Financial Dashboard
 - Financial Health Analysis
 - Debt-to-Income Calculation
 - EMI Ratio Calculation
 - Settlement Prediction
 - Negotiation Letter Generator
 - AI Recommendations
 - Responsive UI
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11. API Endpoints

Authentication

- POST /auth/register
- POST /auth/login

Dashboard

- GET /dashboard/dashboard-data

Loans

- GET /loans
- POST /loans
- PUT /loans/{id}
- DELETE /loans/{id}

Financial Health

- GET /financial-health

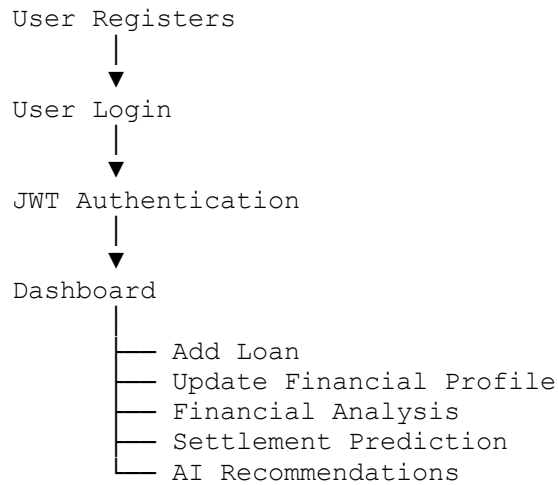
Settlement

- POST /settlement/predict

AI

- POST /ai/recommendations
-

12. Project Workflow



13. Advantages

- Easy financial monitoring
 - AI-powered suggestions
 - Secure authentication
 - Responsive design
 - Simple user interface
 - Fast performance
 - Real-time calculations
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14. Future Enhancements

- Credit score integration
 - Bank API integration
 - Payment reminders
 - Mobile application
 - Multi-language support
 - PDF report generation
 - Machine Learning prediction
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15. Conclusion

FinRelief AI provides an intelligent platform for debt management and financial recovery. It combines FastAPI, React, and AI-driven financial analysis to help users make informed financial decisions, improve repayment planning, and better manage debt.

This documentation is suitable for a **college mini/major project**, internship submission, or GitHub project README. It can also be expanded into a **25–40 page IEEE-style project report** if your college requires a full report.

give more matter

Yes. Below is a **detailed project documentation** for your **FinRelief AI** project. This version is suitable for a **major project report (30–50 pages)** after adding screenshots.

FINRELIEF AI

AI-Powered Debt Relief & Financial Recovery Platform

CHAPTER 1: INTRODUCTION

1.1 Introduction

Financial debt has become one of the major challenges faced by individuals due to increasing living expenses, personal loans, education loans, credit cards, and business liabilities. Many users are unable to understand their financial condition or plan their repayments effectively.

FinRelief AI is a smart web-based application developed to help users analyze their financial condition, calculate debt burden, predict settlement opportunities, and receive AI-powered financial guidance. The system combines modern web technologies with financial analytics to provide users with clear insights into their financial health.

The application securely stores user information and generates reports based on income, expenses, outstanding loans, and EMI commitments. It also provides recommendations to improve financial stability and reduce debt stress.

1.2 Problem Statement

Many people face difficulties in managing multiple loans and understanding their financial condition. Existing financial applications mainly focus on transaction tracking and do not provide intelligent debt analysis or settlement recommendations.

Problems include:

- Difficulty tracking multiple loans.
- Lack of financial health analysis.
- No debt settlement prediction.
- Limited financial guidance.
- Poor visualization of debt status.

FinRelief AI addresses these problems through intelligent analysis and AI-based recommendations.

1.3 Objectives

The major objectives are:

- Develop a secure financial management platform.
 - Provide debt analysis.
 - Calculate EMI burden.
 - Calculate Debt-to-Income Ratio.
 - Predict loan settlement possibilities.
 - Generate negotiation letters.
 - Provide AI-powered financial suggestions.
 - Improve users' financial planning.
-

1.4 Scope

The project can be used by:

- Individual borrowers
- Salaried employees
- Students
- Small business owners
- Financial consultants
- Banking institutions (future enhancement)

Future improvements include:

- Credit score integration
 - Bank account synchronization
 - Mobile application
 - Real-time notifications
 - Machine learning prediction
 - Investment planning
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CHAPTER 2: LITERATURE SURVEY

Several financial management systems are available, but they mainly provide expense tracking and budgeting. Few applications offer complete debt analysis with AI-based recommendations.

Traditional systems:

- Expense Tracker
- Banking Applications
- Loan EMI Calculators
- Budget Planners

Limitations:

- No financial stress analysis
- No settlement prediction
- No AI recommendations
- No financial health score

FinRelief AI overcomes these limitations.

CHAPTER 3: SYSTEM ANALYSIS

Existing System

Existing systems mainly provide:

- Expense recording
- Budget management
- EMI calculation

Disadvantages:

- No debt prioritization
 - No financial score
 - No AI support
 - No settlement prediction
-

Proposed System

FinRelief AI provides:

- Secure login
- Loan management
- Financial health dashboard
- Debt analysis
- Settlement prediction
- AI recommendation engine
- Negotiation letter generation

Advantages:

- User friendly
 - Fast calculations
 - Accurate financial analysis
 - Responsive interface
 - Secure authentication
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CHAPTER 4: SYSTEM REQUIREMENTS

Hardware Requirements

- Intel Core i3 or above
- 4 GB RAM
- 10 GB Storage
- Internet Connection

Recommended:

- Intel Core i5
 - 8 GB RAM
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Software Requirements

Operating System

- Windows 10/11

Programming Languages

- Python
- JavaScript

Frontend

- React.js
- HTML5
- CSS3
- Axios

Backend

- FastAPI
- SQLAlchemy
- Pydantic
- JWT

Database

- SQLite
- PostgreSQL

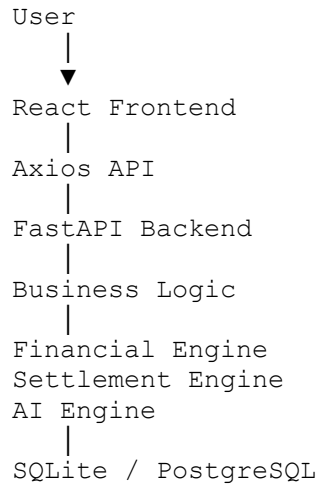
Deployment

- GitHub
- Render

Development Tools

- VS Code
- Git
- Postman

CHAPTER 5: SYSTEM ARCHITECTURE



CHAPTER 6: MODULES

User Authentication

Features

- Register
- Login
- JWT Token
- Password Encryption

Benefits

- Secure access
 - User-specific data
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Dashboard

Displays

- Monthly Income
- Monthly Expenses
- Monthly Surplus
- Outstanding Debt
- EMI Ratio
- Financial Health Score
- Financial Stress

Loan Management

Users can

- Add loans
- Edit loans
- Delete loans
- Track outstanding balance
- View EMI

Financial Health Analysis

Calculates

- Debt-to-Income Ratio
- EMI Ratio
- Monthly Surplus
- Financial Health Score
- Financial Stress Level

Settlement Predictor

Predicts

- Settlement Amount
- Settlement Percentage
- Recovery Possibility
- Settlement Recommendation

AI Recommendation Module

Provides

- Spending advice
- Loan repayment order
- EMI reduction tips

- Savings suggestions
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Negotiation Letter Generator

Automatically creates professional settlement request letters that users can submit to banks.

History Module

Stores

- Previous loans
 - Financial analysis
 - Settlement reports
-

Know Your Rights

Provides

- Consumer rights
 - Debt recovery guidelines
 - Legal awareness
-

CHAPTER 7: DATABASE DESIGN

User Table

Field	Type
user_id	Integer
username	String
email	String
password	String

Loan Table

Field	Type
loan_id	Integer
user_id	Integer
lender	String
loan_amount	Float
outstanding_amount	Float
interest_rate	Float
emi	Float

Financial Profile Table

Field	Type
profile_id	Integer
user_id	Integer
monthly_income	Float
monthly_expenses	Float
existing_debts	Float
lump_sum_available	Float
financial_health_score	Float

CHAPTER 8: API ENDPOINTS

Authentication

- POST /auth/register
- POST /auth/login

Dashboard

- GET /dashboard/dashboard-data

Loans

- GET /loans
- POST /loans
- PUT /loans/{id}
- DELETE /loans/{id}

Financial Health

- GET /financial-health

Settlement

- POST /settlement/predict

AI

- POST /ai/recommendations
-

CHAPTER 9: ADVANTAGES

- Secure authentication
 - Easy loan management
 - AI-powered financial analysis
 - Fast calculations
 - Responsive design
 - Professional dashboard
 - Easy deployment
 - Scalable architecture
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CHAPTER 10: APPLICATIONS

- Personal finance management
 - Banking sector
 - Financial consulting
 - Loan recovery agencies
 - Educational institutions
 - FinTech startups
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CHAPTER 11: LIMITATIONS

- Manual data entry
- Internet connection required
- No direct bank integration
- Limited to registered users

CHAPTER 12: FUTURE ENHANCEMENTS

- Machine Learning prediction
- Credit score integration
- Mobile application
- UPI payment support
- SMS reminders
- Email notifications
- PDF report generation
- Voice assistant
- Multi-language support
- Banking API integration

CHAPTER 13: TESTING

Testing performed:

- Unit Testing
- Integration Testing
- API Testing
- Authentication Testing
- Database Testing
- UI Testing
- Deployment Testing

CHAPTER 14: CONCLUSION

FinRelief AI is an intelligent financial recovery platform that helps users understand their financial position, manage loans, and make informed repayment decisions. The system combines React, FastAPI, SQLAlchemy, JWT authentication, and AI-driven analysis to provide a secure, scalable, and user-friendly solution. It supports better financial planning through dashboard insights, settlement prediction, and personalized recommendations, making it valuable for individuals seeking improved debt management and long-term financial stability.